

Dr. Amareshwara Sainadh Chamarthi

CONTACT INFORMATION

Division of Engineering & Applied Science,
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RESEARCH INTERESTS

- Physics-constrained algorithms for GPU-accelerated high-fidelity CFD.
- Low-dissipation numerical schemes for high-speed compressible flows and digital twins.
- Hybrid physics-based and data-driven methods for simulation acceleration.
- Multi-species, multiphase, and real-gas hypersonic flow simulation.

TECHNICAL SKILLS

- **Languages:** Fortran (primary), Python, C, MATLAB
- **Frameworks:** SciPy optimization, custom high-order CFD solvers, NVIDIA Warp
- **Parallel Computing:** MPI, OpenACC for NVIDIA GPU architectures

OPEN-SOURCE SOFTWARE

- **WA-CR-Warp** — Wave-appropriate GPU solver for 3D compressible flows implemented in NVIDIA Warp. Validated on the inviscid Taylor-Green Vortex at 512^3 on a single A100 80 GB GPU. <https://github.com/aschamarthi/WA-CR-Warp>.
- **Wave-Multiphase** — Wave-appropriate GPU solver for compressible multicomponent and multiphase flows implemented in NVIDIA Warp.
<https://github.com/aschamarthi/Wave-Multiphase>
Papers related to the codes are published in Journal of Computational Physics

APPOINTMENTS

Postdoctoral Fellow, California Institute of Technology, USA

Sep 2023 to Current

- Developed a physics-appropriate discretization framework that has been used for simulations of a wide class of problems from hypersonic flows, multiphase, combustion, etc., without empirical tuning parameters. The approach directly takes into consideration the wave structure associated with eigenvectors of the concerned PDE. The concerned work is published as a series of papers in the Journal of Computational Physics.
- Wave-appropriate reconstruction framework treats acoustic, entropy, and vortical characteristic waves according to their distinct physical properties, minimizing dissipation on smooth flow features while preserving stability near shocks; the optimized nonlinear N th-order schemes match or outperform standard linear $(N + 2)$ th-order schemes at no additional cost.
- Eliminated the contact discontinuity detector via a rank-1 algebraic correction along the entropy right eigenvector, reducing solver wall-time by 29–41% across benchmark configurations at no accuracy penalty. Algorithms are implemented in NVIDIA Warp framework with significant speedup and scaling to millions of grid points.
- Reduced scheme optimization from four empirical parameters requiring ~ 100 surrogate-assisted CFD evaluations (Bayesian optimization) to a single scalar ($\eta_a^* = 0.54$) in ~ 25 evaluations; the wave-appropriate design recovers from first principles what Bayesian and deep reinforcement learning approaches rediscover empirically, transferring without retuning from subsonic turbulence to Mach 10 flows.
- Demonstrated that second-order physics-consistent schemes suppress spurious vortices in shear-layer flows on coarser grids than higher-order Bayesian-optimized and deep reinforcement learning schemes.
- Conducted real-gas DNS of hypersonic boundary layers (with Dr. Bellan, JPL/Caltech); demonstrated that perfect-gas simulations over-predict heat transfer by 20–30%, critical for thermal protection system design.

- Built and deployed a high-performance Fortran CFD solver accelerated with MPI and OpenACC for GPU architectures; secured competitive HPC allocations on NASA Ames (NAS) and TACC through successful research proposals.

Postdoctoral Fellow, Israel Institute of Technology, Israel,

Jul 2019 to Sep 2023

- Designed advanced low-dissipation shock-capturing schemes that outperform state-of-the-art ENO/TENO-type methods in turbulence-resolving accuracy at fixed grid resolution, as demonstrated across a wide variety of problems in peer-reviewed publications.
- Identified that spectral properties of viscous flux discretization critically affect stability in compressible flows involving shocks and turbulence; this connection, previously overlooked in the literature, was found to be essential for hypersonic boundary layer transition simulations.
- Built a high-performance Fortran-based numerical solver for high-speed flows, accelerated using MPI and OpenACC for parallel computing on GPU architectures.
- Supervised three PhD students directly; officially co-advisor for one, with a focus on high-speed multicomponent flow simulations.
- Trained personnel at Rafael and Elbit Systems in the use of the developed solver and delivered invited technical talks at both organizations.

Scientist, Fluid Mechanics and Thermal Division, VSSC, Indian Space Research Organization,
Oct 2018 to Jun 2019

Scientist, PRG, VSSC, Indian Space Research Organization,

Feb 2010 to Mar 2015

- Designed and analyzed solid rocket motors for the Crew Escape System of India's Human Space Programme; designs were successfully flight tested in 2019.
- Developed performance evaluation codes for solid rocket motors, including shock-boundary layer interaction and nozzle flow separation models.

EDUCATION

Ph.D., Aeronautics and Astronautics, University of Tokyo, Tokyo, 2018,

- Developed high-order numerical schemes and preconditioners for strongly anisotropic PDEs on general meshes, with applications to magnetized plasma in spacecraft propulsion systems.
- Introduced a first-order hyperbolic formulation for anisotropic diffusion equations, enabling robust discretization in regimes where standard elliptic solvers fail.
- Supported by a Japanese Government MEXT scholarship.

M.Tech., Aerospace Engineering, Indian Institute of Technology, Bombay, 2008,

B.Tech., Jawaharlal Nehru Technological University, Hyderabad, 2006.

AWARDS

- Technion academic excellence fellowship, Technion, Israel.
- Japanese Government scholarship (**MEXT**), University of Tokyo
- MHRD scholarship by Government of India, IIT Bombay

JOURNAL PUBLICATIONS

16. **Amareshwara Sainadh Ch.,** “Wave-Appropriate Reconstruction of Compressible Multiphase and Multicomponent Flows: Fully Conservative and Semi-Conservative Eigenstructures”, Journal of Computational Physics, 2026.

15. **Amareshwara Sainadh Ch.**, “*Wave-appropriate reconstruction of compressible flows: physics-constrained acoustic dissipation and rank-1 entropy wave correction*”, Journal of Computational Physics, 2026.
14. R Raj, S Saini, N Rao Vadlamani, **Amareshwara Sainadh Ch.**, "Assessment of Gradient-based Reconstruction and Artificial Diffusivity Methods in Simulating High-Speed Compressible Flows", Computers and Fluids, 2026.
13. **Amareshwara Sainadh Ch.**, “*Physics Appropriate Interface Capturing Reconstruction Approach for Viscous Compressible Multicomponent Flows*”, Computers & Fluids, 2025.
12. **Amareshwara Sainadh Ch.**, “*Wave-Appropriate Multidimensional Upwinding Approach for Compressible Multiphase Flows*”, Journal of Computational Physics, Vol. 538, October 2025.
11. Natan Hoffmann, **Amareshwara Sainadh Ch.**, Steven H. Frankel, “*Centralized Gradient-Based Reconstruction for Wall Modelled Large Eddy Simulations of Hypersonic Boundary Layer Transition*”, Journal of Computational Physics, Vol. 512, September, 2024.
10. **Amareshwara Sainadh Ch.** “*Efficient high-order Gradient-based Reconstruction for compressible flows*”, Journal of Computational Physics, Vol. 486, August 2023.
9. **Amareshwara Sainadh Ch.**, Natan Hoffmann, Steven H. Frankel, “*A Wave Appropriate Discontinuity Sensor Approach for Compressible Flows*”, Physics of Fluids, 2023.
8. Hemanth Chandra Vamsi, **Amareshwara Sainadh Ch.**, Natan Hoffmann, Steven H. Frankel, “*On the Application of Gradient Based Reconstruction for Flow Simulations on Generalized Curvilinear and Dynamic Mesh Domains*”, Computers & Fluids, 2023.
7. **Amareshwara Sainadh Ch.**, Natan Hoffmann, Hiroaki Nishikawa, Steven H. Frankel, “*Implicit gradients based novel conservative numerical scheme for compressible flows*”, Journal of Scientific Computing, 95(1), p.17, 2023.
6. **Amareshwara Sainadh Ch.**, Hemanth Chandra Vamsi, Natan Hoffmann, Sean Bokor, Steven H. Frankel, “*On the role of spectral properties of viscous flux discretization for flow simulations on marginally resolved grids*”, Computers & Fluids, Vol. 251, January 2023.
5. **Amareshwara Sainadh Ch.** “*Gradient Based Reconstruction: Inviscid and viscous flux discretizations, shock capturing, and its application to single and multicomponent flows*”, Computers & Fluids, Vol. 250, January 2023.
4. **Amareshwara Sainadh Ch.**, Sean Bokor, Steven H. Frankel, “*On the importance of high-frequency damping in high-order conservative finite-difference schemes for viscous fluxes*”, Journal of Computational Physics, Vol. 460, 2022.
3. **Amareshwara Sainadh Ch.**, Steven H. Frankel, “*High-Order Central-Upwind shock capturing scheme using a Boundary Variation Diminishing (BVD) Algorithm*”, Journal of Computational Physics, Vol. 427, 2021.
2. **Amareshwara Sainadh Ch.**, H. Nishikawa, K. Komurasaki, , “*First order hyperbolic approach for anisotropic diffusion equation*”, Journal of Computational Physics, Vol. 396, pp. 243-263, 2019.

1. **Amareshwara Sainadh Ch.**, K. Komurasaki, R. Kawashima, “*High-order upwind and non-oscillatory approach for steady state diffusion, advection-diffusion and application to magnetized electrons*”, Journal of Computational Physics, Vol. 374, pp. 1120 -1151, 2018.

INVITED TALKS

7. **Amareshwara Sainadh Ch.**, *Physics-appropriate numerical methods for high-speed compressible flow simulations*, Texas Tech , December 2025.
6. **Amareshwara Sainadh Ch.**, *Physics-appropriate numerical methods for high-speed compressible flow simulations*, University of Cincinnati, June 2025.
5. **Amareshwara Sainadh Ch.**, *Efficient wave-appropriate numerical methods for high-speed compressible flow*, California Institute of Technology, April 2025.
4. **Amareshwara Sainadh Ch.**, *Efficient and physics consistent numerical methods for high-speed flows*, Iowa State University, February 2025.
3. **Amareshwara Sainadh Ch.**, *Efficient and accurate numerical methods for high-speed compressible flows*, Louisiana State University, February 2024.
2. **Amareshwara Sainadh Ch.**, *How to use gradients efficiently and accurately in simulations*, IMI Joint Usage Research Program, Kyushu University, November 2023
1. **Amareshwara Sainadh Ch.**, *Importance of Spectral Properties of Viscous Flux Discretization and Gradient-Based Reconstruction Approach*, Computers & Fluids seminars, <https://doi.org/10.52843/c>

GRADUATE STUDENT SUPERVISION

Co-advisor, PhD student, Technion

- Co-advisor of Sean Bokor for his PhD on high-speed multicomponent flow simulations using gradient based reconstruction Jul 2021 – Present

PROFESSIONAL SERVICE

Reviewer:

- Journal of Computational Physics
- Computers & Fluids
- Engineering with Computers
- International Journal for Numerical Methods in Fluids
- Physics of Fluids
- Computers and Mathematics with Applications

CONFERENCES

24. Kumar, Ritu R., Nagabhushana R. Vadlamani, and **Amareshwara Sainadh Ch.**. *Effect of Wall Temperature on Pressure Fluctuations Beneath High-Speed Boundary Layers*. 27th AIAA International Space Planes and Hypersonic Systems and Technologies Conference. 2026.
23. Bokor, Sean, **Amareshwara Sainadh Ch.**, Natan Hoffmann, and Steven Frankel. *Numerical Simulations of Supersonic Multi-Component and Reacting Flows: Towards Combustion in a Supersonic Cavity*. AIAA Scitech Forum and Exposition, 2026.
22. **Amareshwara Sainadh Ch.**, Josette Bellan, *Real-Gas Direct Numerical Simulation of a Single-Species and Binary-Species Hypersonic Boundary Layer with Heat Transfer*, AIAA Scitech Forum and Exposition, 2026.
21. Natan Hoffmann, **Amareshwara Sainadh Ch.**, Steven H. Frankel. *Wall-Modeled Large-Eddy Simulation of Hypersonic Boundary Layer Transition over a Cone-Cylinder-Flare*, AIAA Scitech Forum and Exposition, 2026.

20. Natan Hoffmann, **Amareshwara Sainadh Ch.**, Steven H. Frankel. *Effect of Wall Temperature on Hypersonic Boundary Layer Transition over a Compression Ramp*, AIAA Scitech Forum and Exposition, 2026.
19. **Amareshwara Sainadh Ch.**, Josette Bellan, *Direct Numerical Simulation of a Boundary Layer Induced by a Hypersonic Fluid Flow over a Wall*, 35th International Symposium on Shock Waves, Brisbane, Australia, 2025.
18. Ritu Raj, Shivam Saini, N R Vadlamani, **Amareshwara Sainadh Ch.**. *Assessment of Local Artificial Diffusivity and Gradient-Based Reconstruction Methods for High-Speed Flows*, 25th International Shock Interaction Symposium, Delft, Netherlands, 2024.
17. Sean Bokor, **Amareshwara Sainadh Ch.**, Steven H. Frankel. *Numerical Simulation of Multi-Component Flows with Variable Thermodynamic Properties Towards Supersonic Flow Over a Cavity*, AIAA Scitech Forum and Exposition, 2024.
16. Hemanth Chandra Vamsi, **Amareshwara Sainadh Ch.**, Steven H. Frankel. *Impinging Wall Curvature Effects on Flow and Screech Tones in Choked Under Expanded Supersonic Jets*, AIAA Scitech Forum and Exposition, 2024.
15. **Amareshwara Sainadh Ch.**, "Adaptive Interface Capturing Approach for Multicomponent Flows", 76th Annual Meeting of the Division of Fluid Dynamics, November 19-21, 2023; Washington, DC.
14. **Amareshwara Sainadh Ch.**, *High resolution numerical methods for CFD*. CFDimpact, Technion, Israel (2023).
13. Natan Hoffmann, **Amareshwara Sainadh Ch.**, Hemanth Chandravamsi K., Steven H. Frankel. *Large Eddy Simulation of Transitional and Turbulent Hypersonic Flow*. AIAA Scitech Forum and Exposition (2023). <https://doi.org/10.2514/6.2023-1383>.
12. Hemanth Chandra Vamsi, **Amareshwara Sainadh Ch.**, Natan Hoffmann, Steven H. Frankel. *GPU-Accelerated Numerical Study of Temperature Effects in Choked Under-Expanded Supersonic Jets*. AIAA Scitech Forum and Exposition (2023). <https://doi.org/10.2514/6.2023-0976>.
11. **Amareshwara Sainadh Ch.**, Steven Frankel. *A family of gradient based reconstruction schemes for compressible Navier-Stokes equations*. 22nd Computational Fluids Conference (2023). Cannes, France.
10. Natan Hoffmann, **Amareshwara Sainadh Ch.**, Steven Frankel. *A Subgrid-Scale Transition Model for Hypersonic Boundary Layer Transition in Wall-Modelled Large Eddy Simulations*. 22nd Computational Fluids Conference (2023).
9. **Amareshwara Sainadh Ch.**, Natan Hoffmann, Steven Frankel, "Shock-capturing for high-order compact schemes via Boundary variation diminishing approach and its applications", 16th US National Congress on Computational Mechanics, 25-29 July, 2021, Chicago, USA
8. Z. Wang, **S. Chamarthi**, R. Kawashima, K. Komurasaki, H. Koizumi, "A One-dimensional Hybrid Particle-Fluid Simulation with Electron Inertia in a Hall Effect Thruster," Asian Joint Conference on Propulsion and Power 2018, Xiamen, China, Mar. 14-17, 2018.
7. **Amareshwara Sainadh Ch.**, Z. Wang, R. Kawashima, K. Komurasaki, "Weighted Nonlinear Schemes for Magnetized Electron Fluid in Quasi-neutral Plasma," 2018 AIAA Aerospace Sciences Meeting, Kissimmee, FL, Jan. 8-12, 2018.

6. R. Kawashima, Z. Wang, **S. Chamarthi**, H. Koizumi, K. Komurasaki, "Hyperbolic System Approach for Magnetized Electron Fluids in ExB Discharge Plasmas," 2018 AIAA Aerospace Sciences Meeting, Kissimmee, FL, Jan. 8-12, 2018.
5. Y. Hamada, R. Kawashima, J. Bak, **S. Chamarthi**, Z. Wang, G. Ito, M. Mizukawa, "Refillable Electric Propulsion with In-situ Produced Fuels in Space," 35th International Electric Propulsion Conference, Atlanta, GA, Oct. 8-12, 2017.
4. **Amareshwara Sainadh Ch.** Jeenu R., Jayaprakash J, "Analysis of large Solid rocket motor Static firing tests", ISTS/IEPC/NSAT Joint Conference, Kobe, Japan, July 6-10, 2015.
3. **Amareshwara Sainadh Ch.**, Jeenu R., Jayaprakash J., Srinivasan V., "Variation of burning rate scale factors in SRM", 9th International High energy material conference, Trivandrum, 2014.
2. **Amareshwara Sainadh Ch.** and Rajkumar S. Pant. "Studies on Morphing Aircraft Design including Engine parameters using Genetic Algorithm", 14th AIAA Aviation Technology, Integration, and Operations Conference, AIAA AVIATION Forum, (AIAA 2014-2723)
1. **Amareshwara Sainadh Ch.**, R Jeenu, J. Jayaprakash, Srinivasan V., S. Ramakrishnan, "Flow Separation in Rocket Motors during Sea Level Static Test", 62nd International Astronautical Congress, Cape town, South Africa, 2011.

POSTER PRESENTATIONS

- **Amareshwara Sainadh Ch.**, K. Komurasaki, R. Kawashima, "High-order schemes for Magnetized Electron Fluid in Quasi-neutral plasma", Proceedings of Space Transportation Symposium, STEP-2017-077, Sagamihara, Kanagawa Japan, January 18-19, 2018.
- **Amareshwara Sainadh Ch.**, K. Komurasaki, R. Kawashima, "Variable specific weight Particle-in-cell model for Hall thruster modelling", UKAREN, Hakodate, Japan, 2016.

TECHNICAL REPORTS

- **Amareshwara Sainadh Ch.**, Jeenu R., "*Grain Design and performance data for the solid motors of Crew Escape System of Human Space Program*," ISRO-VSSC-TR-0102-0-13, **RESTRICTED**, Trivandrum, India, February 2013.
- **Amareshwara Sainadh Ch.**, Khadar Voli K., Jeenu R., "*Performance prediction of Solid Rocket Motor Igniters*," ISRO-VSSC-TR-0008-0-14, **RESTRICTED**, Trivandrum, India, December 2013.
- **Amareshwara Sainadh Ch.**, Kiran P., "*Correlation between Pressure and Thrust Oscillations in S200 Solid Rocket Motor*," LVM3-ITR-02-SC-10, Trivandrum, India, December 2011.
- **Amareshwara Sainadh Ch.**, Jeenu R., "*Flow separation studies in large solid rocket motors*," GNDD/11/2011, Trivandrum, India